

Missed Benchmarks: Predicting the 2030 Emissions Gap

Executive Summary

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INTRODUCTION

Scientists have stated that the tipping point into ecological and environmental devastation is a 2°C increase in the average global temperature compared to the pre-industrial baseline. They have also strongly urged that we try to avoid even the 1.5°C threshold, as its devastating effects may take centuries to recover from. To stay below the 1.5°C limit, countries would have to reduce the amount of greenhouse gases (GHGs) they emit each year, working to avoid a global concentration limit set by scientists.

It is understood that annual GHG emissions have accelerated exponentially across the globe since the pre-industrial period (often considered by scientists to be 1850 – 1900). In 1992, the United Nations drafted its Framework Convention on Climate Change (UNFCCC) to devise an agreement to limit the increase in GHGs in the atmosphere. This was built upon by the Kyoto Protocol, adopted in 1997. Each successive treaty was more aggressive and urgent in its goals. However, we continued to see rising atmospheric GHG concentrations and greater yearly emissions during that time. We also saw accelerated effects of climate change.

This is the basis for the Paris Agreement, a treaty adopted in December 2015 by 197 United Nations countries pledging to fight climate change. Each party submitted emission reduction goals in Intended Nationally Determined Contributions (INDCs) before and Nationally Determined Contributions (NDCs) after its adoption.

ABSTRACT

Paris Agreement signatories had to submit plans for how and by how much they would lower their GHG emissions through 2020, 2030, and 2050. Such plans - INDCs and NDCs for pre- and post-Agreement plans, respectively - would be reviewed and revised at each benchmark based on the nations' achievements toward their GHG target. In a ramp-up style, the Agreement requires countries to recommit to more aggressive plans with each consecutive iteration. While NDCs include more than just GHG emission reduction targets, they will be the focus of this project.

The data for this project contains historical GHG emissions from all countries and the benchmarks laid out in their INDCs and NDCs. The trajectory of the historical data was extrapolated forward to 2030 to provide projections of GHG emissions to compare to the goals of the Paris Agreement. The difference was measured as the emissions gap. The analysis conducted with this data focused on historical global greenhouse gas emissions and identifying the largest contributors to greenhouse gas emissions, both historically and currently. The countries of focus were identified based on their behavior as top emitters in 2015. The top 10 emitters in 2015 are a primary focus of this analysis.

This project aimed to examine how well the parties that signed the Paris Agreement are meeting their goals. The data reflects that, on the aggregate, countries that have signed onto the Paris Agreement are not doing enough so far to limit warming below 2°C. The largest emitters have not enacted policy changes that put them on track to meet the goals of their NDCs. These nations are also not expected to reach peak emissions before 2030. One goal of the Paris Agreement is to peak as soon as possible; the goal in 2015 was for peak emissions by 2025. The longer it takes for emissions to peak, the more difficult it will be to reach net zero goals and stay on track to limit warming.

ANALYSIS

When attempting to lower global GHG emissions, it is important to know which countries have the greatest impact. A few patterns are noticeable when evaluating cumulative historical emissions from 1850 to 2020:

Ten countries have the greatest cumulative emissions over the period (in descending order): the United States, China, Russia, India, Germany, the United Kingdom, Japan, France, Ukraine, and Brazil. Figure 2 shows the share of global emissions each of these ten countries make up each year, with all other countries grouped together as “All Others”, from 1950 – 2020.

In this period, these ten countries accounted for roughly 50% or more of yearly global emissions. The widening or narrowing of each country’s area on the graph reflects increases or decreases in yearly emissions over time.

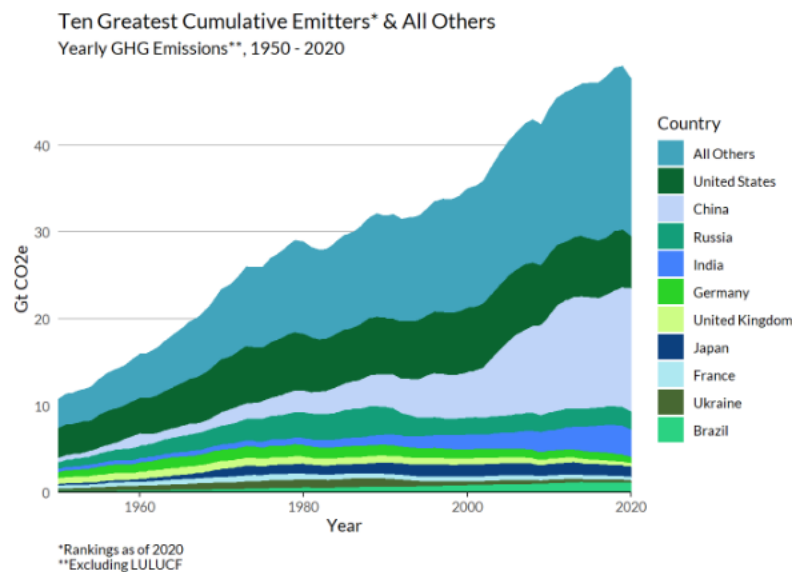


Figure 1

The United States saw its most significant changes from the 1870s to the early 1970s. As one of the first countries to industrialize and liberalize its economy, the composition of the United States's economic activity has not seen major change in the last half-century. We see this reflected in the graph: Over the last 40 years, the United States's yearly emissions have increased slightly but have held relatively consistently.

Fellow early-industrializers that made the list include France, Germany, and the United Kingdom. By contrast, China, and India, to a lesser extent, show increases over the same period, rising significantly around the turn of the 21st century. These increases in emissions reflect the countries' efforts to rapidly industrialize in recent decades and open their economies to global markets.

Population plays an essential role as well. If this data were represented in emissions per capita, it could significantly change the composition of these rankings. A large share of these countries' emissions is likely, in part, due to their large populations.

The combination of early industrialization, a large economy, and a large population provide the foundation for the United States being the largest single contributor to Earth's greenhouse gas concentrations since 1850.

Table 1. Top 10 Cumulative Emitters, 1850 – 2020

Country	Total excl. LULUCF summed	Total incl. LULUCF summed	LULUCF summed
United States	557.11	592.42	35.31
China	363.95	426.10	62.15
Russia	175.73	186.76	11.03
India	119.19	149.72	30.54
Germany	115.40	114.59	-0.81
United Kingdom	99.21	102.52	3.31
Japan	76.41	75.43	-0.98
France	59.75	56.54	-3.21
Ukraine	51.48	57.27	5.79
Brazil	49.03	130.26	81.23

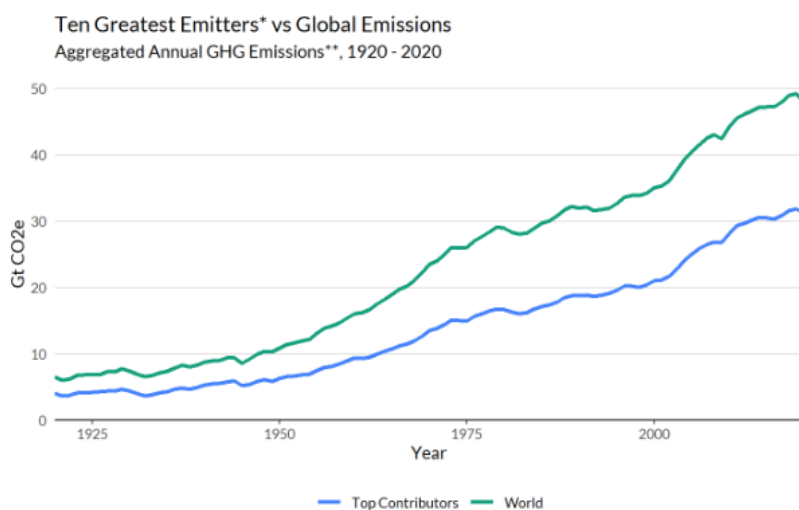
The nations of focus for the projections were determined by the highest emissions at the time the Paris Agreement was signed and NDCs were being crafted. Comparing the largest cumulative emitters since 1850 to the largest emitters of 2015, we see some countries appear on both lists.

In 2015, the ten countries that emitted the most GHGs were, in descending order: China, the United States, India, Russia, Japan, Brazil, Iran, Germany, Indonesia, and Saudi Arabia. This effect is noticeable in the data – aggregated historical emissions of those ten mirrors the global emissions data nearly perfectly since 1900.

Since 1980, this group of countries has constituted roughly the same percentage of global emissions each year, reflecting the massive effect these countries have on the aggregate.

Table 2. 2015 Top Emitters

Country	Total excl. LULUCF summed	Total incl. LULUCF summed	LULUCF summed
China	12.79	11.94	-0.85
United States	6.78	6.05	-0.74
India	2.92	2.62	-0.30
Russia	2.06	1.46	-0.61
Japan	1.32	1.27	-0.06
Brazil	1.17	1.39	0.22
Iran	0.93	0.95	0.02
Germany	0.90	0.88	-0.02
Indonesia	0.83	1.48	0.65
Saudi Arabia	0.75	0.74	-0.01



*In 2015

**Excluding LULUCF

Top Contributors: China, United States, India, Russia, Japan, Brazil, Iran, Germany, Indonesia, and Saudi Arabia

Figure 2

The Top Contributors submitted INDCs in the fall of 2015 as part of their intention to sign the Paris Agreement at the UNFCCC's twenty-first session of the Conference of the Parties (COP 21) that November. Historical data for the sum of the Top Contributors and the global total were used to predict each grouping's yearly emission trajectory for 2021 through 2030.

First, a BAU line, reflecting behavior before 2016, was predicted to estimate emissions if pre-INDC policies were maintained. Next, 2015 to 2020 served as each group's post-INDC and NDC policy behavior period.

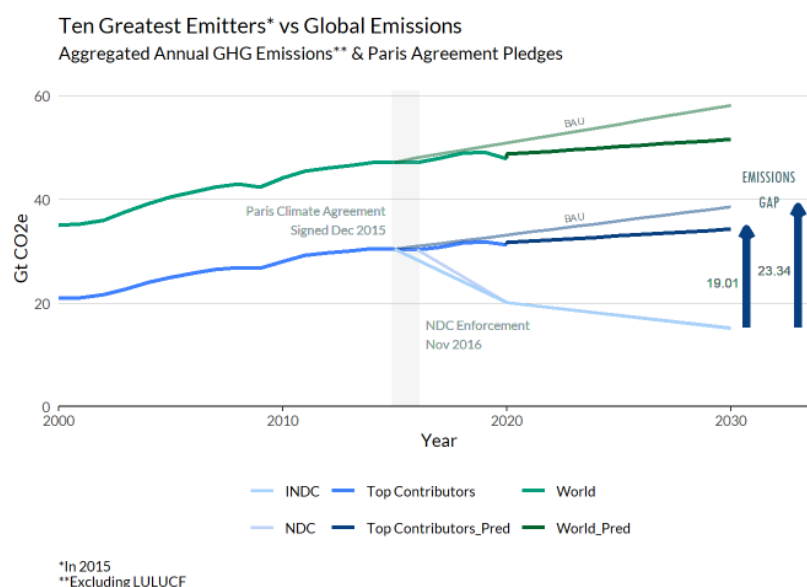


Figure 3

In Figure 4, the collective pre-2020 goals for the Top Contributors can be seen on the INDC and NDC lines at the year 2020. This provides a connection between their behavior at the time they were making their post-2020 pledges and their shortcomings of meeting their pre-2020 pledges. The steepness of the line reflects the degree of change they would need to make to be on track. Prior to COP 26, parties were requested to submit new or updated NDCs that represent a progression in intention and actions compared to their first NDCs.

For the years 2016 to 2020, global emissions, and those of the Top Contributors, were lower than their BAU predictions were estimated to be. However, the decrease in yearly emissions came nowhere close to as low as needed to reach 2020 pledges or be on track to reach 2030 pledges.

When aggregating the pledges for the period ending in 2020, their yearly emissions were meant to be lowered to 20.06 Gt of CO₂ equivalent by 2020. The collected data from that year shows that the actual aggregate was more than 10 Gt greater than the goal.

The INDC and NDCs these countries submitted were significantly more ambitious than their pre-2020 targets: 15.15 and 15.25 Gt CO₂e before 2030, respectively.¹ However, estimates show that emissions will continue to increase from 2020 to 2030.

Additionally, the gap between actual emissions and the pledges will be even larger in 2030 than that for 2020. Compared to the aggregated NDCs for the Top Contributors, estimates put emissions for the year

¹ Russia is the only country that submitted an NDC with a less aggressive goal than its INDC, from a maximum effort of 75 Gt CO₂e to 70 Gt CO₂e.

2030 at 19.02 Gt CO₂ equivalent higher than pledged. This is, however, smaller than the gap between pledged emissions and BAU emissions for the year 2030.

As noted above, our Top Contributors collectively failed to meet their pre-2020 pledges and are on track to fail to meet the pledges of their current NDCs. Plans will need to be considerably more aggressive in the next round of NDCs and actions must be more urgent throughout the decade if members hope to close the gap.

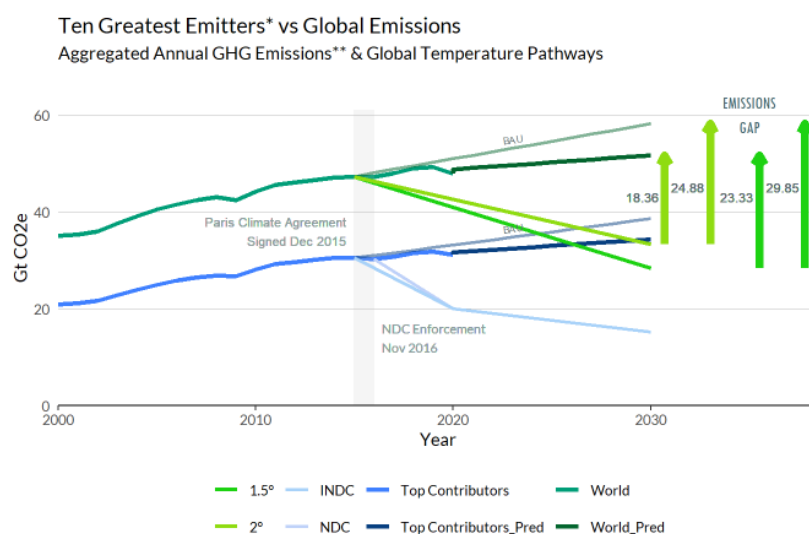


Figure 4

As previously stated, the main goal of the Paris Agreement is to limit global warming to 2°C while making best efforts to stay below 1.5°C compared with pre-industrial levels. In its Fifth Assessment Report², the IPCC predicts that the paths most likely to occur require 72% and 49.5% reductions in global Like with the NDCs, there is a large gap between the pathways and predicted global emissions in the year 2030. However, like the Top Contributors data, the predicted emissions through 2030 are lower than those predicted for the BAU trajectory. As mentioned, the emissions gap here is significant.

Table 3. Goal Emissions, Predicted Emissions, and Predicted Emissions Gaps in 2030

Group	Pathway	Emissions (Gt)	Surplus from Goal (Gt)
World	1.5°	28.28	
World	2°	33.26	
Top Contributors	NDC	15.25	
World	Predicted	51.61	18.35, 23.33
Top Contributors	Predicted	34.26	19.01
World	BAU	58.14	24.88, 29.86
Top Contributors	BAU	38.59	23.34

CONCLUSION

The Montreal Protocol was created in 1987 to reverse the depletion of the ozone layer in the atmosphere by phasing out substances that cause ozone depletion. In 2016, scientists confirmed that the ozone layer was recovering, with the possibility of a full return by 2066. The international cooperation from this treaty

² This report was the basis for the Paris Agreements; the Sixth Assessment Report was released this year and will be the basis for the next round of NDCs.

is an example of a successful collective, multilateral response to a global climate issue. These countries can make these GHG emission changes happen. However, so far, no treaty has had much success in its attempt to significantly curb GHG emissions and the impacts of climate change.

Governments and corporations must address and combat climate change through policies that reduce GHG emissions. These policies require innovative technologies, changing the status quo, and massive investments in both. Transitions to renewable energy away from fossil fuel combustion is a great first step. However, there must be a multi-pronged approach across industries, sectors, and levels of government. Many developing nations will require international financial support to make these transitions.

Governments often step in to fill gaps where corporations see no benefit to an action. This is where strong pro-environmental policies and regulations come in. Still, corporations must see the benefit of moving to green, sustainable actions and technologies. Voters in the former case and consumers in the latter hold power to encourage those changes. We must fight for those who will be hurt first and greatest, and time is running out.